IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-DS-002

1. Topic (12 words max)

Interactive Sinhala Text Reader for the Visually Impaired Using OCR, NLP, AI.

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	N/A
Year	N/A

5. Brief description of the research problem including references (300 words max) – references not included in word count.

• **Nature / Gravity of the Problem:** Blind and partially sighted Sinhala readers lack an affordable, hands-free way to independently access information, such as physical newspaper content, school textbooks, article, notices and more. This places a limit on daily information access and may also be applied to illiterate people, who have the ability to speak and understand Sinhala but not to read.

• **Impact of the Problem:** The problem leads to social exclusion, as a group of people will be excluded with latest ongoing information and highlights current limits Sinhala-language assistive technology. It violates equal access to information and showcases poor accessibility support in print media.

• **FoC Research Cluster and Industry Verticals:** The project falls under the *Autonomous Intelligent Machines and Systems* research cluster, serving assistive technology, media & publishing, healthcare, and education industries.

• **Sustainable Development Goals (SDGs):** The problem aligns with the UN education standards SDG 4(Quality Education), 9 (Industry, Innovation and Infrastructure), 10 (Reduced Inequalities), and 11(Sustainable Cities and Communities) by promoting inclusive education (4), affordable innovation (9), reduced inequalities (10), and accessible public information (11).

6. Brief description of Existing Research and Systems (300 words max)

Summary of existing products/systems:

Currently, wearable assistive devices like OrCam, MyEye and Envision Glasses provide OCR and text-to-speech applications for visually challenged users. However, these are out of the budget of the average Sri Lankan user, very often exceeding $2,000, and have limited/zero support for the Sinhala language. Though mobile applications such as Google Lookout to read physical documents like newspapers, their use requires holding a phone, with no hands-free operation option available.

Summary of existing research:

Most of the academic research in this domain is focused on the development of better Sinhala OCRs, such as using Tesseract, Surya or custom LSTM models, and basic Text-to-Speech (TTS) conversion. While many studies have focused on the digitization of documents, most fail to handle complex, multi-column layouts & fonts seen in Sri Lankan Content. In addition, most current studies are simple, linear "read aloud" systems without interactivity, such as querying text or asking for summaries, receiving a personalized output.

Gaps/Limitations identified:

1. **Language & Layout Support:** Existing tools struggle with the complexities of Sinhala documents (columns, headers, and mixed fonts), often reading across headliners incorrectly.

2**. Lack of Interactivity:** Current systems are static; they read text from start to finish. There is a gap in systems that allow users to have a two-way dialogue (e.g., "Summarize this article" or "Skip to the sports section") using LLMs, particularly Sinhala.

3. **Unnatural Audio**: Standard Sinhala Text-to-Speech (TTS) engines often sound robotic and lack natural intonation, causing listener fatigue.

4. **Cost**: High-end wearable solutions are not affordable for the local demographic.

References

- **OrCam MyEye**: OrCam Technologies. (n.d.). *OrCam MyEye - The Ultimate Visual Aid Device*. Retrieved from https://www.orcam.com
- **Envision Glasses**: Envision. (n.d.). *Envision Glasses: AI-powered Smart Glasses for the Blind*. Retrieved from https://www.letsenvision.com/glasses

- **Google Lookout:** Google LLC. (2023). *Lookout - Assisted Vision. Android Play Store. Retrieved from* https://play.google.com/store/apps/details?id=com.google.android.apps.accessibility.reveal
- **Sinhala OCR Research**: Anuradha, I., Liyanage, C., & Weerasinghe, R. (2020). *Estimating the Effects of Text Genre, Image Resolution and Algorithmic Complexity needed for Sinhala Optical Character Recognition.* University of Colombo School of Computing. https://www.researchgate.net/publication/353710137_Estimating_the_Effects_of_Text_Genre_Image_Resolution_and_Algorithmic_Complexity_needed_for_Sinhala_Optical_Character_Recognition
- **Deep Learning OCR**: Ambegoda, T. (2025). *Benchmarking OCR Models for Sinhala and Tamil Document* Digitization.https://www.researchgate.net/publication/391058761_Benchmarking_OCR_Models_for_Sinhala_and_Tamil_Document_Digitization

Data Collection Strategy:

- **News Articles:** Scraped daily from official portals like *Mawbima* (www.mawbima.lk) and Lankadeepa (www.lankadeepa.lk) to build a verified knowledge base.
- **Textbooks:** Digital PDF copies obtained from the *Educational Publications Department* (www.edupub.gov.lk) for syllabus verification.
- **Font Files (.ttf)**: Make use of .ttf files to generate dataset for each font and to train on it.
- **Manual Dataset Creation:** Create a dataset of images of various documents needed for final testing phases.
- **OCR Training:** Synthetic character datasets will be sourced from open repositories like Kaggle (*Sinhala Synthetic OCR Dataset*) to augment manual scans of physical newspapers. Sinhala Synthetic OCR Dataset (https://www.kaggle.com/datasets/ransakaravihara/sinhala-synthetic-ocr-dataset)

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

• How the solution would be applied

The application first captures snapshots of physical documents such as newspapers or school textbooks held by the user using the glasses-mounted camera. These images are processed to extract maximum textual and structural information, including layout-aware Sinhala OCR and metadata generation.

Headliners, Topic and Titles are extracted through the Topological Sinhala Character reader component for accurate and faster access to the subject of the text, which implement the use of Medial Axis Transform (MAT). This information is passed along with the text of the body into the second component, which is a content-aware post-OCR error correction model which work on Sinhala & OCR specific mistakes and corrects them based on context provided.

Using a ranked matching retrieval mechanism, the system attempts to identify and retrieve the most relevant stored version of the scanned document from a local unstructured database, followed by official publisher websites and other trusted sources if needed. The compiled and matched content is then passed into a Retrieval-Augmented Generation (RAG) pipeline integrated with an LLM, which complies data available interactive use. Users can ask questions, request summaries, or have content explained in a guided manner through voice interaction.

The system also maintains a personalized profile that adapts responses based on user behavior and preferences, converting raw speech into structured queries to deliver the most relevant and natural Sinhala audio output.

• Physical Application

From a physical perspective, the solution consists of a pair of glasses with an attached camera connected via USB to a nearby laptop, which receives a live visual feed for processing. The user interacts with the system using the laptop's microphone and speakers, with future extensions supporting fully wireless glasses with built-in audio components.

• Addressing the Identified Gaps:
The proposed system addresses these gaps by supporting complex Sinhala newspaper and textbook layouts through zone detection, enabling natural two-way voice interaction using LLMs, providing more natural Sinhala speech output, and offering a low-cost glasses-plus-PC solution suitable for Sri Lankan users.

• System Perspective (Input–Process–Output):
The system takes physical newspaper or School textbook images and Sinhala voice input as **inputs**, processes them using layout detection, Sinhala OCR, multi-source retrieval, and RAG-based LLM reasoning, and delivers **outputs** as accurate, summarized, and interactive Sinhala audio responses.

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

How can members gain essential specialized domain expertise?

- Members are actively reviewing IEEE/ACM research papers on low-resource language OCR and RAG architectures to understand the current state.
- To handle the complexities of the Sinhala script (modifiers and complex characters), the team will study the linguistic principles demanded for proper tokenization and text-to-speech functionality.
- Members can also get real time technical updates as research into Sinhala plus Artificial Intelligence is a trending and active topic with frequent updates.
- The team will study WCAG (Web Content Accessibility Guidelines) to ensure that the voice interaction design can actually be used by the visually impaired.

How can a member gain the technical knowledge and skills required for the project?

- Member 1- (Component 1): The member assigned to OCR is studying OpenCV, Tesseract and Surya related documents as well as results, specifically related to large and font applied texts, and is studying into theory about Medial Axis Transform.
- Member 2 (Component 4): Actively exploring current available TTS & STT models (Hugging Face – Whisper) and their available documentation. Checked availability of current research papers on personalization of LLMs, specifically in the subject of Sinhala.
- Member 3 (Components 2): Looking into the current rule-based mechanisms, specifically research done by the UOM, as well as the technical knowledge behind transformer-based models.
- Member 4 (Component 3): Explore the research papers and study RAG pipelines, vector databases like ChromaDB/Pinecone, and Sinhala LLMs like SinLLaMA. Learn multi-source retrieval, web scraping with BeautifulSoup/Selenium, and dynamic text adaptation.
- Hardware Integration: The team is experimenting with IoT protocols to establish low-latency communication between the smart glasses' camera and the laptop processing unit.

How is the data required for the project collected?

Primary Data (Physical):

- The primary data would be collected based on the component, with them mainly being
  - Various Font Templates to create characters of each font
  - Documents to test and generate data required for the auto correction module
  - Current available database on newspapers, schoolbooks, article and more

Secondary Data (Digital):

- Secondary data would be collected and generate as required such as synthetic datasets for fonts with augmentation applied to improve the accuracy of component 1 as well as train the 2nd component. Web scraping and access to news and other API's and large information sets will be required for component 3. For the voice module, more Sinhala audio with inbuilt noise, accents and other complications might become useful.

Note – Due to the large diversity and complexity of possible sources, the study will be done on a thoughtfully identified set of Sinhalese publications that are either newspapers, schoolbooks or articles that are available.

Does this project require ethical clearance?

The project involves visually impaired human users, voice interactions, and potential collection of image and audio data, which may include personal or sensitive information.

Although no medical intervention is involved and formal ethical approval is not required, ethical awareness is essential to ensure informed consent, data privacy, secure handling of recordings, and responsible use of AI-generated content.

9. Objectives and Novelty

Main Objective

To design and develop a low-cost, glasses-based assistive system that enables visually impaired users to read, understand, and interact with Sinhala newspapers through hands-free image capture, intelligent retrieval, and natural voice-based interaction.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22072238	Topological Sinhala Character Reader	Apply the Mathematical concepts of Medial Axis Transform (MAT) to find the geometric centerline of Large Font Text such a Headline, topics and titles and create a more robust OCR technique that specializes in above type of text.	MAT has not been systematically applied to Sinhala headlines or display typography as a primary recognition mechanism. Existing OCR pipelines treat skeletonization as obsolete; this work repositions MAT as a font-invariant technique for extreme typography. The method targets non-body text, a category largely ignored by most OCR research.

IT22004468	Adaptive Conversational Personalization and Voice Interaction Module	Adaptive Conversational Personalization and Voice Interaction Module	The Adaptive Conversational Personalization and Voice Interaction Module oversees processing user voice input, identifying user intent, monitoring user interaction behavior, adapting the level of explanation and content structure in explanations, and generating personalized Sinhala speech output that is continually changing in accordance with user preferences and comprehension.
IT22259134	Context-Aware Sinhala post-OCR error correction model	Prepare a clean Sinhala corpus, normalize it, and split into sentences. Generate noisy-clean pairs by adding realistic OCR style errors, then train a transformer model to correct OCR text. Compare against simple baselines and evaluate using CER/WER with a few real OCR examples.	Targets Sinhala OCR-specific correction (similar letters/diacritics) using context-aware transformers, not only rules. Includes a sinhala oriented OCR-noise dataset design for reliable training and evaluation in an assistive reading.
IT22120670	Advanced Document retrieval and Information generation	Implement multi-source retrieval, vector DB context building, RAG for Sinhala content understanding, and dynamic simplify/elaborate adaptation.	Integration of RAG-driven dynamic text simplification and elaboration within document retrieval pipelines; persistent multi-source knowledge base evolution; and real-time semantic expansion tailored for Sinhala language processing

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Lakmini	Abeywardhana	
Co-Supervisor	Ms.	Adya	Dissanayake	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				